In [3]:
```
import pandas as pd
import numpy as np
```

In [4]:
```
df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")

df.head()
```

Out[4]:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Educat
0	41	Yes	Travel_Rarely	1102	Sales	1	
1	49	No	Travel_Frequently	279	Research & Development	8	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	
3	33	No	Travel_Frequently	1392	Research & Development	3	
4	27	No	Travel_Rarely	591	Research & Development	2	

5 rows × 35 columns

In [5]:
```
# Number of rows and columns
print("Dataset size:", df.shape)

# Column names and data types
df.info()

# Basic numerical summary
df.describe()
```

```
Dataset size: (1470, 35)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64
 19  MonthlyRate               1470 non-null   int64
 20  NumCompaniesWorked        1470 non-null   int64
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64
 24  PerformanceRating         1470 non-null   int64
 25  RelationshipSatisfaction  1470 non-null   int64
 26  StandardHours             1470 non-null   int64
 27  StockOptionLevel          1470 non-null   int64
 28  TotalWorkingYears         1470 non-null   int64
 29  TrainingTimesLastYear     1470 non-null   int64
 30  WorkLifeBalance           1470 non-null   int64
 31  YearsAtCompany            1470 non-null   int64
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

Out[5]:

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount	Em
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	
mean	36.923810	802.485714	9.192517	2.912925	1.0	
std	9.135373	403.509100	8.106864	1.024165	0.0	
min	18.000000	102.000000	1.000000	1.000000	1.0	
25%	30.000000	465.000000	2.000000	2.000000	1.0	
50%	36.000000	802.000000	7.000000	3.000000	1.0	
75%	43.000000	1157.000000	14.000000	4.000000	1.0	
max	60.000000	1499.000000	29.000000	5.000000	1.0	

8 rows × 26 columns

In [6]:
```
# Count missing values in each column
df.isnull().sum()
```

Out[6]:
```
Age                         0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
YearsWithCurrManager        0
dtype: int64
```

In [7]:
```
# Count duplicate rows
print("Duplicate rows:", df.duplicated().sum())
```

```
Duplicate rows: 0
```

In [11]:
```
# Create a copy so the original data remains unchanged
clean_df = df.copy()

# Remove duplicate rows
clean_df = clean_df.drop_duplicates()
```

In [12]:
```
cleaning_report = pd.DataFrame({
    "Cleaning Step": [
        "Original number of rows",
        "Original number of columns",
        "Missing values before cleaning",
        "Duplicate rows before cleaning",
        "Missing values after cleaning",
        "Duplicate rows after cleaning",
        "Final number of rows",
        "Final number of columns"
    ],
    "Result": [
        df.shape[0],
        df.shape[1],
        int(df.isnull().sum().sum()),
        int(df.duplicated().sum()),
        int(clean_df.isnull().sum().sum()),
        int(clean_df.duplicated().sum()),
        clean_df.shape[0],
        clean_df.shape[1]
    ]
})

display(cleaning_report)

cleaning_report.to_csv("data_cleaning_report.csv", index=False)
```

	Cleaning Step	Result
0	Original number of rows	1470
1	Original number of columns	35
2	Missing values before cleaning	0
3	Duplicate rows before cleaning	0
4	Missing values after cleaning	0
5	Duplicate rows after cleaning	0
6	Final number of rows	1470
7	Final number of columns	35

In [13]:
```
clean_df.to_csv("cleaned_employee_attrition.csv", index=False)

print("Cleaned dataset saved successfully.")
```

```
Cleaned dataset saved successfully.
```